Activity -6 — Fuel Consumption and Route Comparison

1. Problem Statement

Develop a Python program to calculate fuel consumption and travel cost for a journey. The program uses distance, vehicle mileage and fuel price to calculate the required fuel and total cost.

2. Objectives

- To calculate fuel required for a journey.
- To calculate the total fuel cost.
- To compare travel costs for different routes.
- To understand Python variables and arithmetic operations.

3. Assumptions

For this example, assume:

- Distance of Route 1 = 100 km
- Distance of Route 2 = 150 km
- Vehicle mileage = 20 km per litre
- Fuel price = ₹100 per litre
- The vehicle mileage and fuel price remain constant.

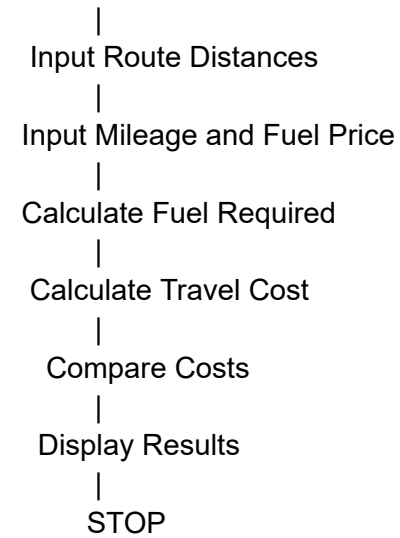
These are sample values for educational purposes.

4. Algorithm

1. Start.
2. Enter the distance of Route 1 and Route 2.
3. Enter vehicle mileage and fuel price.
4. Calculate the fuel required for each route.
5. Calculate the fuel cost for each route.
6. Compare the total costs.
7. Display the fuel requirements and costs.
8. Stop.

5. Flowchart

START



6. Pseudocode

```
START
INPUT distance1
INPUT distance2
INPUT mileage
INPUT fuel_price

IF any input is invalid
   PRINT "Invalid input"
ELSE
   fuel1 = distance1 / mileage
   fuel2 = distance2 / mileage

   cost1 = fuel1 * fuel_price
   cost2 = fuel2 * fuel_price

   PRINT fuel1, cost1
   PRINT fuel2, cost2

   IF cost1 < cost2
      PRINT "Route 1 is cheaper"
   ELSE IF cost2 < cost1
      PRINT "Route 2 is cheaper"
   ELSE
      PRINT "Both routes cost the same"
   ENDIF
ENDIF

STOP
```

7. Python Source Code

File name: `activity6.py`

```
distance1 = float(input("Enter Route 1 distance (km): "))
distance2 = float(input("Enter Route 2 distance (km): "))
mileage = float(input("Enter vehicle mileage (km/litre): "))
fuel_price = float(input("Enter fuel price per litre: "))

if distance1 <= 0 or distance2 <= 0 or mileage <= 0 or fuel_price <= 0:
    print("Invalid input")

else:
    fuel1 = distance1 / mileage
    fuel2 = distance2 / mileage

    cost1 = fuel1 * fuel_price
    cost2 = fuel2 * fuel_price

    print("Route 1 fuel required:", round(fuel1, 2), "litres")
    print("Route 1 travel cost: Rs.", round(cost1, 2))

    print("Route 2 fuel required:", round(fuel2, 2), "litres")
    print("Route 2 travel cost: Rs.", round(cost2, 2))

    if cost1 < cost2:
        print("Route 1 is cheaper")
    elif cost2 < cost1:
        print("Route 2 is cheaper")
    else:
        print("Both routes cost the same")
```

8. Test Cases

Route 1	Route 2	Mileage	Fuel Price	Expected Result
100 km	150 km	20 km/litre	₹100	Route 1: ₹500; Route 2: ₹750
80 km	120 km	20 km/litre	₹100	Route 1: ₹400; Route 2: ₹600
100 km	150 km	0 km/litre	₹100	Invalid input

9. Output Screenshot and Conclusion

Sample Output:

```
Enter Route 1 distance (km): 100
```

Enter Route 2 distance (km): 150
Enter vehicle mileage (km/litre): 20
Enter fuel price per litre: 100
Route 1 fuel required: 5.0 litres
Route 1 travel cost: Rs. 500.0
Route 2 fuel required: 7.5 litres
Route 2 travel cost: Rs. 750.0
Route 1 is cheaper

Output Screenshot: Run `activity6.py` in Python IDLE or VS Code. Capture the inputs and output, and save the screenshot as `activity6_output.png`.

Conclusion: This program calculates fuel consumption and travel expenses for two routes and identifies the cheaper route using Python arithmetic and conditional statements.

Input code:

```
distance1 = float(input("Enter Route 1 distance (km): "))
distance2 = float(input("Enter Route 2 distance (km): "))
mileage = float(input("Enter vehicle mileage (km/litre): "))
fuel_price = float(input("Enter fuel price per litre: "))

if distance1 <= 0 or distance2 <= 0 or mileage <= 0 or fuel_price <= 0:
    print("Invalid input")

else:
    fuel1 = distance1 / mileage
    fuel2 = distance2 / mileage

    cost1 = fuel1 * fuel_price
    cost2 = fuel2 * fuel_price

    print("Route 1 fuel required:", round(fuel1, 2), "litres")
    print("Route 1 travel cost: Rs.", round(cost1, 2))

    print("Route 2 fuel required:", round(fuel2, 2), "litres")
    print("Route 2 travel cost: Rs.", round(cost2, 2))

    if cost1 < cost2:
        print("Route 1 is cheaper")
    elif cost2 < cost1:
        print("Route 2 is cheaper")
    else:
        print("Both routes cost the same")
```

OUTPUT:

```
Enter Route 1 distance (km): 100
Enter Route 2 distance (km): 150
Enter vehicle mileage (km/litre): 20
Enter fuel price per litre: 100
Route 1 fuel required: 5.0 litres
Route 1 travel cost: Rs. 500.0
Route 2 fuel required: 7.5 litres
Route 2 travel cost: Rs. 750.0
Route 1 is cheaper

========================================
Enter Route 1 distance (km): 80
Enter Route 2 distance (km): 120
Enter vehicle mileage (km/litre): 20
Enter fuel price per litre: 100
Route 1 fuel required: 4.0 litres
Route 1 travel cost: Rs. 400.0
Route 2 fuel required: 6.0 litres
Route 2 travel cost: Rs. 600.0
Route 1 is cheaper

========================================
Enter Route 1 distance (km): 100
Enter Route 2 distance (km): 150
Enter vehicle mileage (km/litre): 0
Enter fuel price per litre: 100
Invalid input
```